Risk / Modeling / Results Report

1. Modeling Approach

Sentiment polarity thresholds (±0.05) were used for classification.

RICE framework was applied for feature prioritization.

Sentiment modeling was conducted using polarity-based classification. The objective was to quantify user satisfaction and identify churn risk signals.

High negative sentiment clusters indicate potential churn risk. If not addressed, these concerns may reduce retention and impact long-term growth metrics. Subjectivity analysis was used to distinguish factual complaints from emotional opinions.

Highly subjective negative reviews were examined to understand emotional drivers, while objective negative reviews were treated as high-priority functional defects. Risk assessment highlights that ignoring recurring negative themes may increase uninstall rates and reduce app store ranking.

2. Risk Assessment

High negative sentiment features may increase churn risk.

Improper rollout of features may reduce retention.

3. A/B Testing Plan

Baseline retention: 22%

Target retention: 27%

Sample size per group: 2152 users.

Two-proportion Z-test with Alpha = 0.05 and Power = 80%.

To validate feature impact, an A/B testing framework was designed. Baseline 28-day retention rate:

22% Target retention rate after feature release: 27% Minimum detectable improvement: 5% A

two-proportion Z-test will be conducted with: Significance Level (Alpha) = 0.05 Statistical Power =

80% Required Sample Size per group = 2152 users Test Duration = 28 days If statistical

significance is achieved, full rollout will proceed. Otherwise, iteration and further experimentation

will be conducted.

4. Expected Results

If statistically significant improvement is observed, feature rollout will proceed.

Otherwise, further iteration and validation will be required

Total Reviews: 2,652 | Average Rating: 4.43 | Positive Sentiment: 77% | Negative Sentiment: 12%